

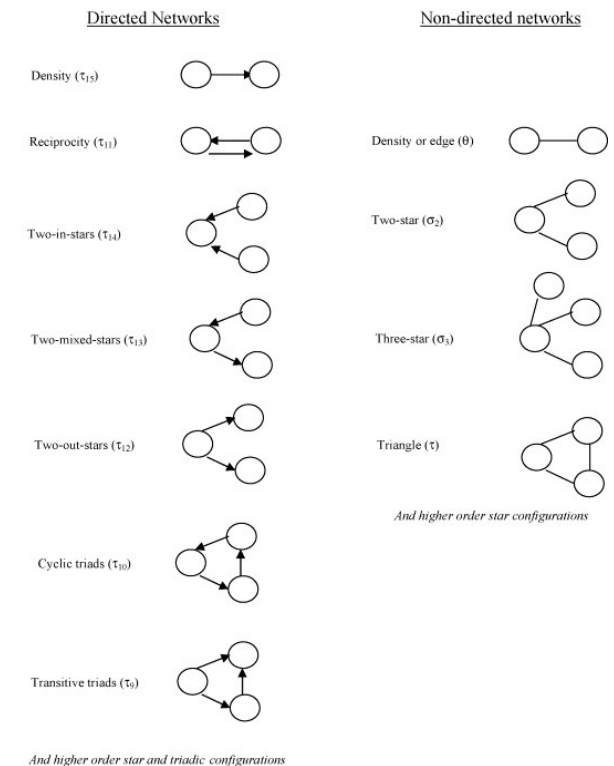
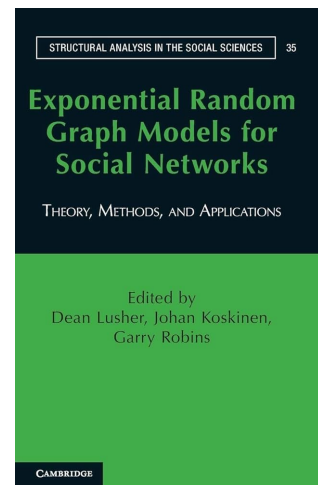
January 2025 | 8.50 am | B-216

# **Social Network Analysis with Statistical Network Models**

Session 3 – Statistical Network Models I

# Session 3 – Statistical network models I

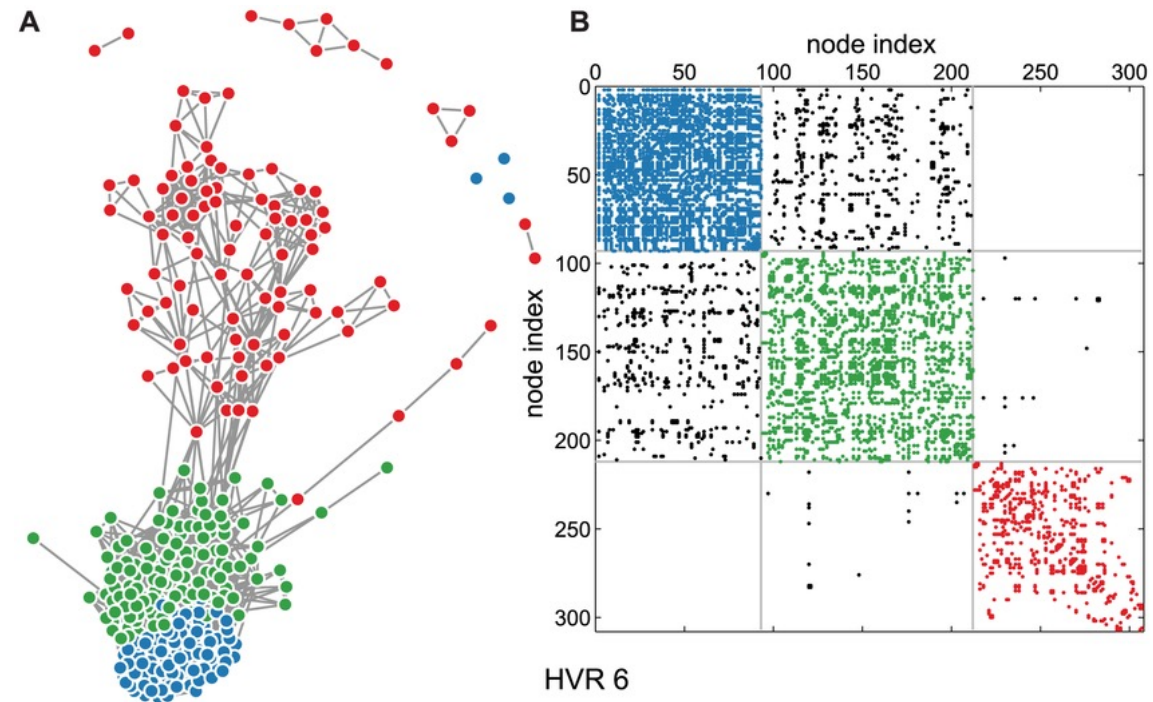
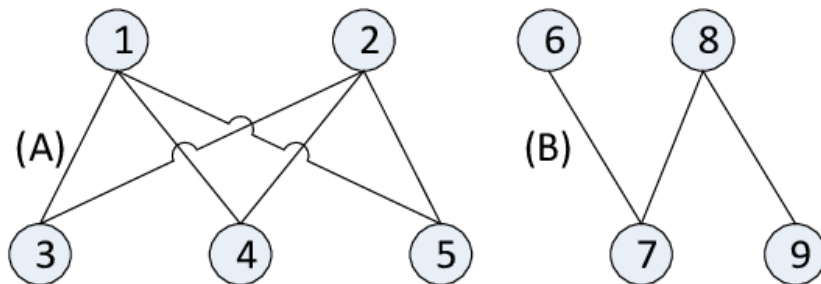
- Model for cross-sectional data, exponential random graph models (ERGMs)
- Methodological background of the models, discussion of link function, estimation procedures etc.



# Block models

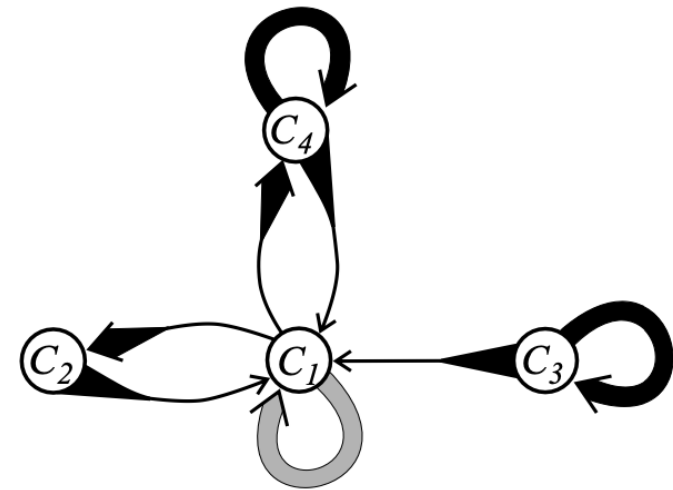
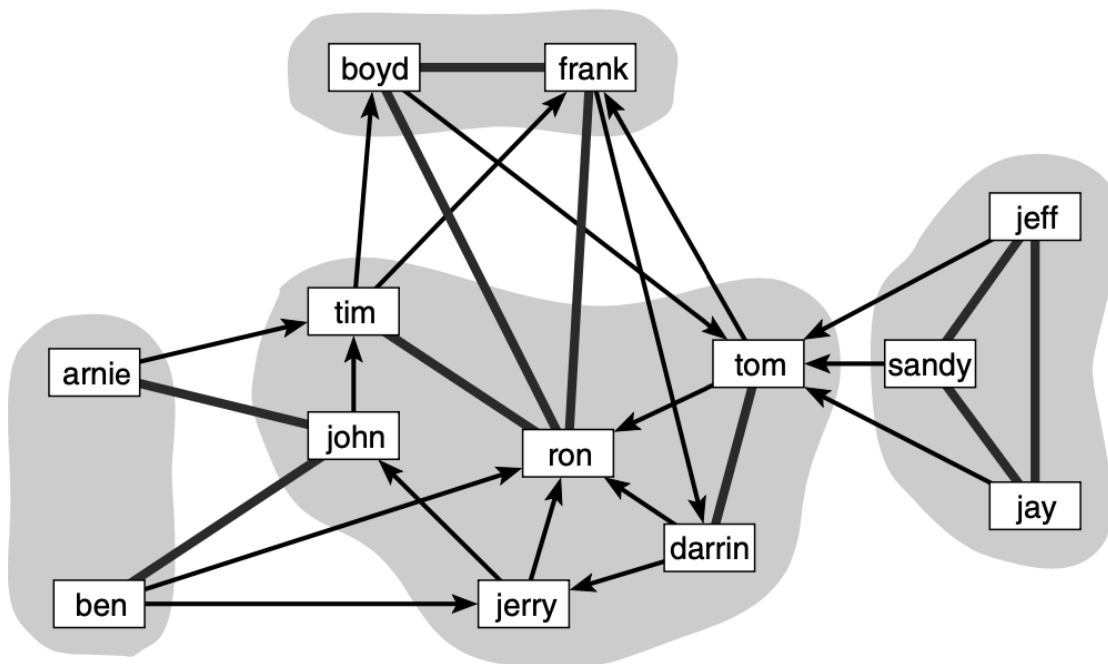
Finding blocks of structurally equivalent nodes:

White, H. C., Boorman, S. A., & Breiger, R. L. (1976). Social structure from multiple networks. I. Blockmodels of roles and positions. *American journal of sociology*, **81**(4), 730-780.



[https://www.researchgate.net/figure/Stochastic-block-modeling-identifies-network-communities-HVR-6-is-shown-in-two-forms\\_fig7\\_257839768](https://www.researchgate.net/figure/Stochastic-block-modeling-identifies-network-communities-HVR-6-is-shown-in-two-forms_fig7_257839768)

## Block models example

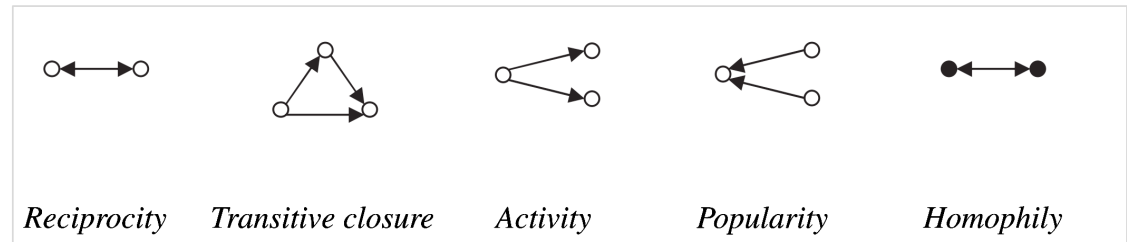


# Statistical network models

- What are the basic ideas behind statistical network models?
- How are dependencies between ties treated in network models, what is the difference compared to (logistic) regressions?
- What are differences of ERGMs in comparison to the methods we discussed last week or other methods you know from your studies?

# ERGMs: model architecture

$$\Pr(X = x) = \frac{e^{\sum_A \eta_A g_A(x)}}{\kappa}$$



- $x$  empirically observed network
- $A$  local structure (e.g. mutual dyad representing reciprocity tendency)
- $g_A(x)$  statistic counting the number of times that  $A$  can be found in  $x$
- $\eta_A$  parameter (to be estimated)
- $\kappa$  correction factor (don't worry about it)

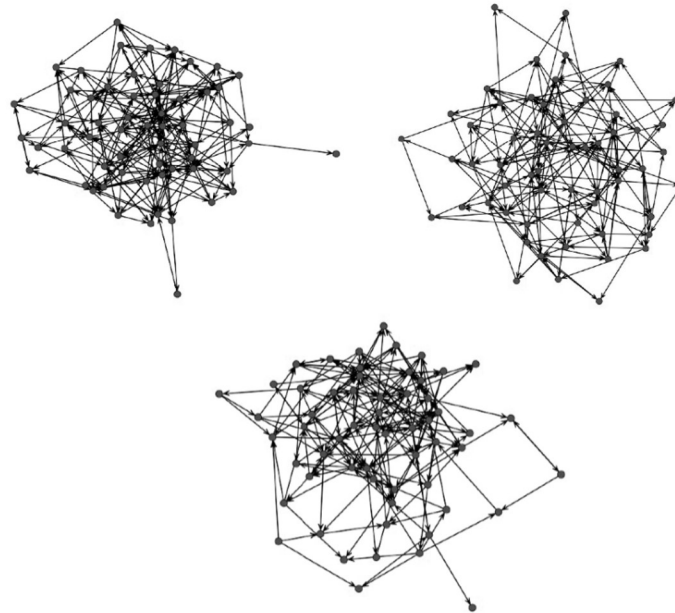
# Intuition behind obtaining parameters

Empirical network



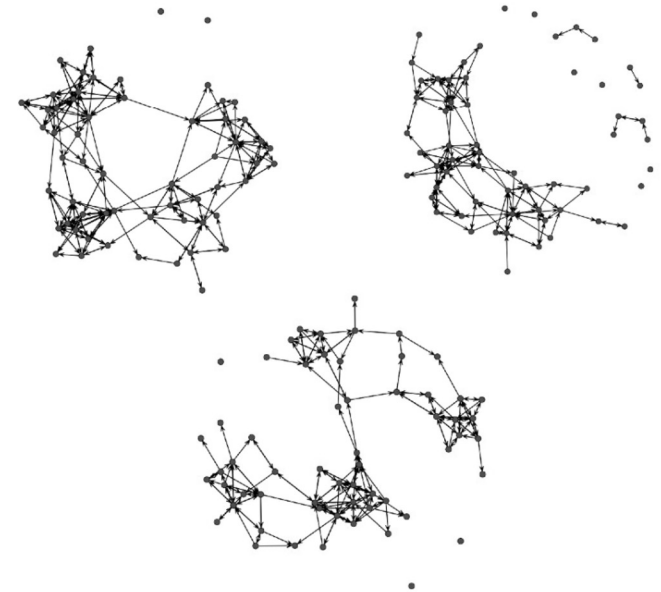
How the actual network looks like

Random networks



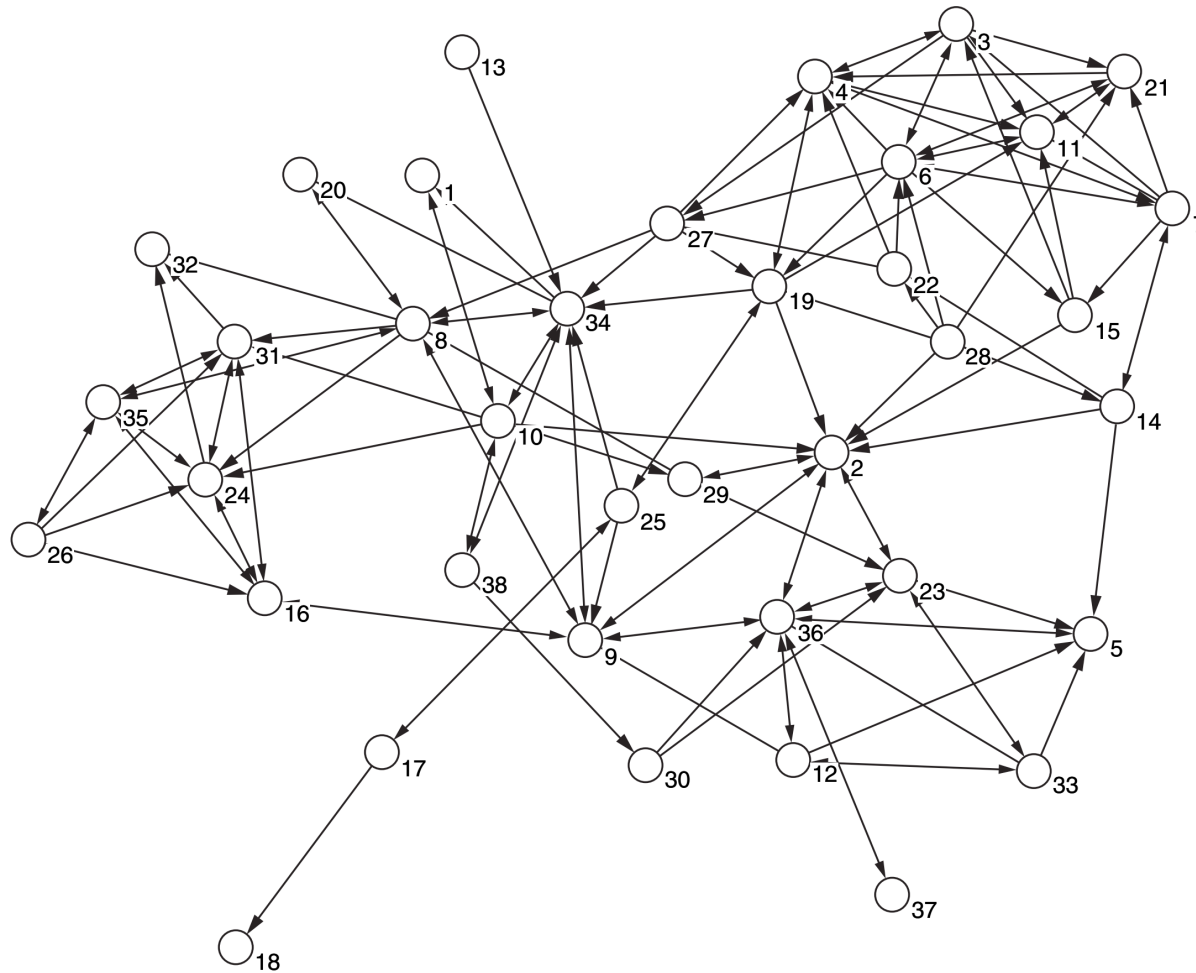
What if ties would be distributed completely at random?

Simulated networks from a simple ERGM

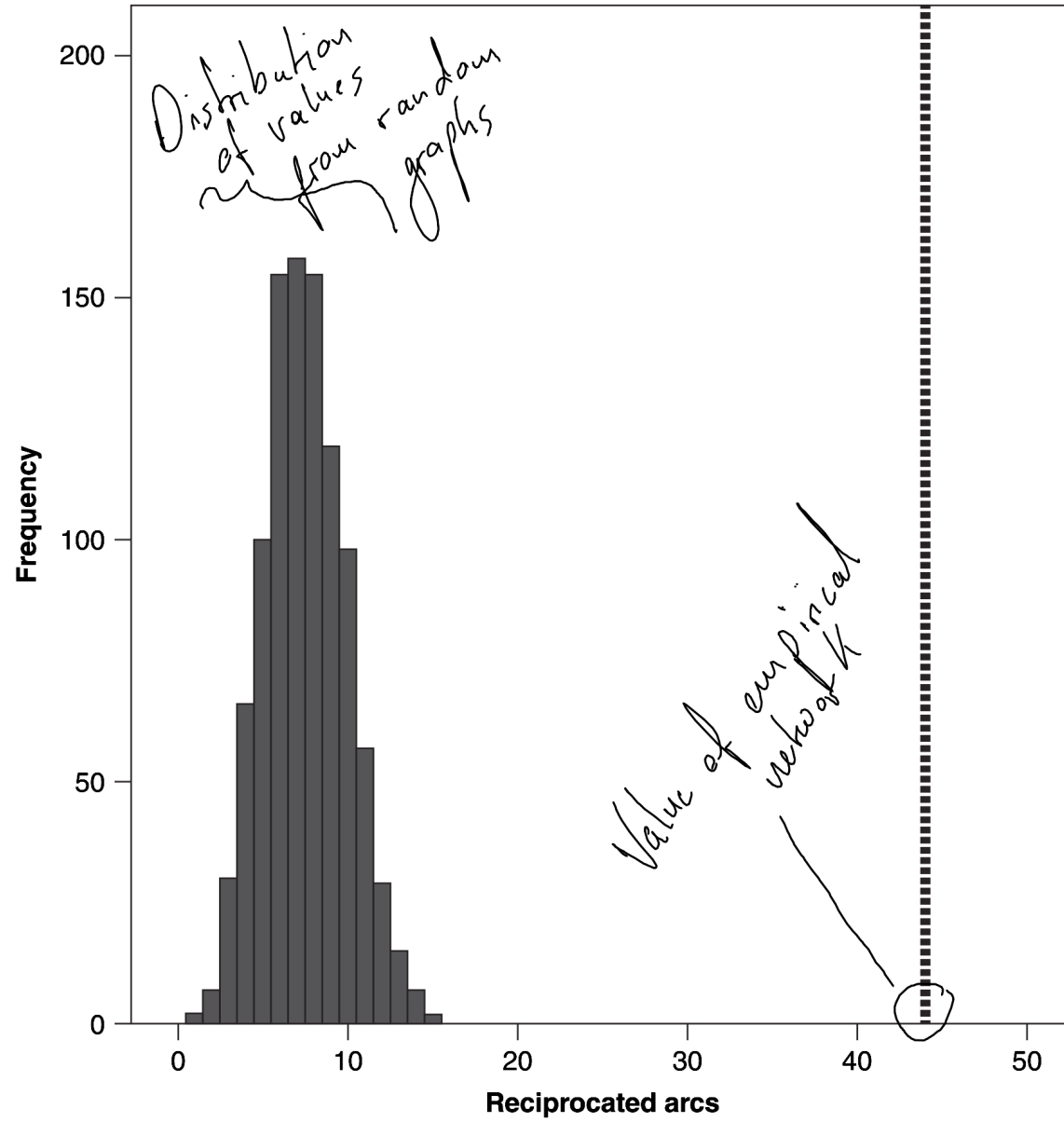


Can we re-build the structure of empirical networks with our model?

# Example: Intra-firm communication network

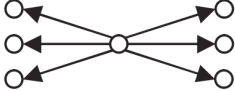
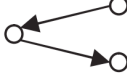
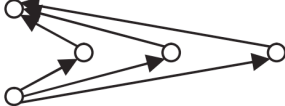
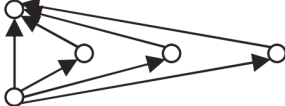
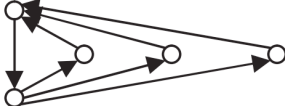


|                   | Random network | Communication network |
|-------------------|----------------|-----------------------|
| Actors            | 38             | 38                    |
| Arcs              | 146            | 146                   |
| Reciprocated arcs | 6              | 44                    |
| Transitive triads | 53             | 212                   |
| In-2-stars        | 292            | 313                   |
| Out-2-stars       | 254            | 283                   |



# Model interpretation

Table 5.1. *ERGM parameter estimates (and standard errors) for communication relations in The Corporation*

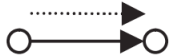
| Network effect                                                |                                                                                      | Estimate (SE) |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------|
| <b>Purely structural effects (endogenous)</b>                 |                                                                                      |               |
| Arc                                                           |    | −1.96 (0.73)* |
| Reciprocity                                                   |    | 2.88 (0.46)*  |
| Popularity (in-degree)                                        |    | −0.27 (0.32)  |
| Activity (out-degree)                                         |    | −0.34 (0.34)  |
| Simple 2-path <sup>3</sup>                                    |   | −0.06 (0.08)  |
| Multiple 2-paths                                              |  | −0.06 (0.09)  |
| Transitivity<br>(transitive path closure of multiple 2-paths) |  | 1.22 (0.19)*  |
| Cyclic closure<br>(cyclic closure of multiple 2-paths)        |  | −0.37 (0.17)* |

### Actor relation effects (exogenous)

(black nodes indicates actor with attribute)

|                                    |                                                                                    |               |
|------------------------------------|------------------------------------------------------------------------------------|---------------|
| Sender (seniority)                 |  | −0.56 (0.29)  |
| Sender (projects)                  |  | 0.01 (0.02)   |
| Receiver (seniority)               |  | 0.08 (0.23)   |
| Receiver (projects)                |  | −0.02 (0.02)  |
| <sup>4</sup> Homophily (seniority) |  | 0.64 (0.26)*  |
| Heterophily (projects)             |  | −0.08 (0.02)* |
| Homophily (office)                 |  | −0.01 (0.17)  |

### Covariate network (exogenous)

|                                    |                                                                                      |              |
|------------------------------------|--------------------------------------------------------------------------------------|--------------|
| Advice entrainment (covariate arc) |  | 1.76 (0.30)* |
|------------------------------------|--------------------------------------------------------------------------------------|--------------|

\* = parameter estimate is greater than two times the standard error in absolute value, indicating the effect is significant (see Section 12.5.1 for details).

# **Introduction to estimation of ERGM parameters**

# Overview

1. Define statistical model, relationship between dependent variable and a vector of independent variables

2. Derivation of estimates for coefficients and standard errors via estimation method, e.g., OLS, ML, MCMC, MoM, etc. → can be frequentist or Bayesian

3. Implementation of estimation procedure in statistical software

4. Estimation of models with empirical data

5. Postestimation: Goodness of fit (GOF), other model diagnostics, additional transformations for interpretation (e.g., average marginal effects)

$$\Pr(X = x|\theta) \equiv P_{\theta}(x) = \frac{1}{\kappa(\theta)} \exp\{\theta_1 z_1(x) + \theta_2 z_2(x) + \dots + \theta_p z_p(x)\}.$$

Joint form of the ERGM

$$L(\theta) = L(\theta; y_1, \dots, y_n) = f(y_1, \dots, y_n; \theta)$$

$$\hat{\theta} = \underset{\theta \in \Theta}{\operatorname{argmax}} L(\theta)$$

e.g., Maximum likelihood



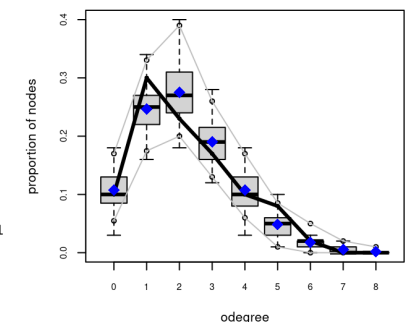
```
Starting maximum pseudolikelihood estimation (MPLE):
Obtaining the responsible dyads.
Evaluating the predictor and response matrix.
Maximizing the pseudolikelihood.
Finished MPLE.
Evaluating log-likelihood at the estimate.
> summary(base_ergm1)
Call:
ergm(formula = network_class1 ~ edges)

Maximum Likelihood Results:

      Estimate Std. Error MCMC % z value Pr(>|z|)
edges  -1.6818    0.1572     0  -10.7  <1e-04 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Null Deviance: 424.2 on 306 degrees of freedom
Residual Deviance: 265.9 on 305 degrees of freedom

AIC: 267.9 BIC: 271.6 (Smaller is better. MC Std. Err. = 0)
```



# Maximum likelihood

$$f(y_1, \dots, y_n; \theta) = \prod_{i=1}^n f(y_i; \theta)$$

$$L(\theta) = L(\theta; y_1, \dots, y_n) = f(y_1, \dots, y_n; \theta)$$

$$\hat{\theta} = \underset{\theta \in \Theta}{\operatorname{argmax}} L(\theta)$$

$$\ell(\theta) = \log L(\theta)$$

$$L(\theta; y) = L(\theta; y_1, \dots, y_n) = \prod_{i=1}^n L(\theta; y_i) = \prod_{i=1}^n f(y_i; \theta)$$

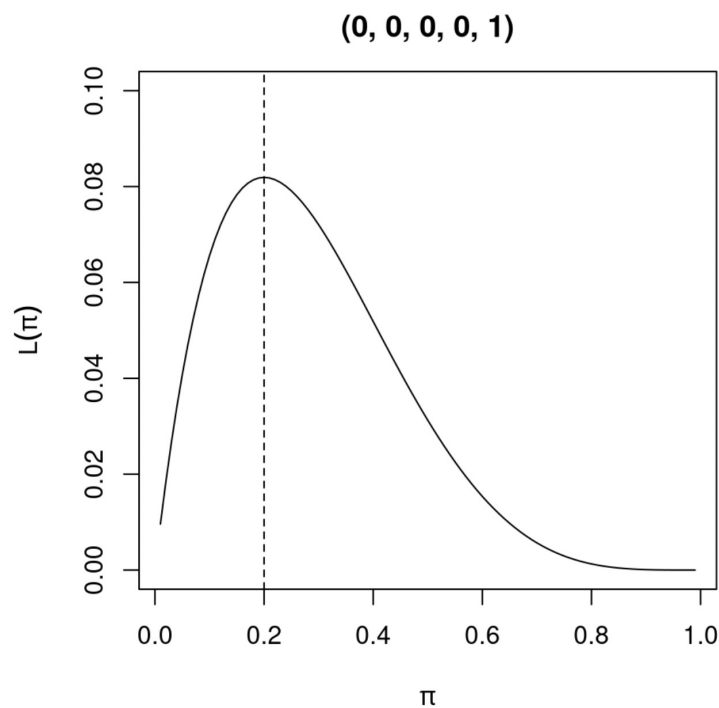
$$\ell(\theta; y) = \ell(\theta; y_1, \dots, y_n) = \sum_{i=1}^n \ell(\theta; y_i) = \sum_{i=1}^n \log f(y_i; \theta).$$

First derivative  
according to theta gives  
us the **score vector**

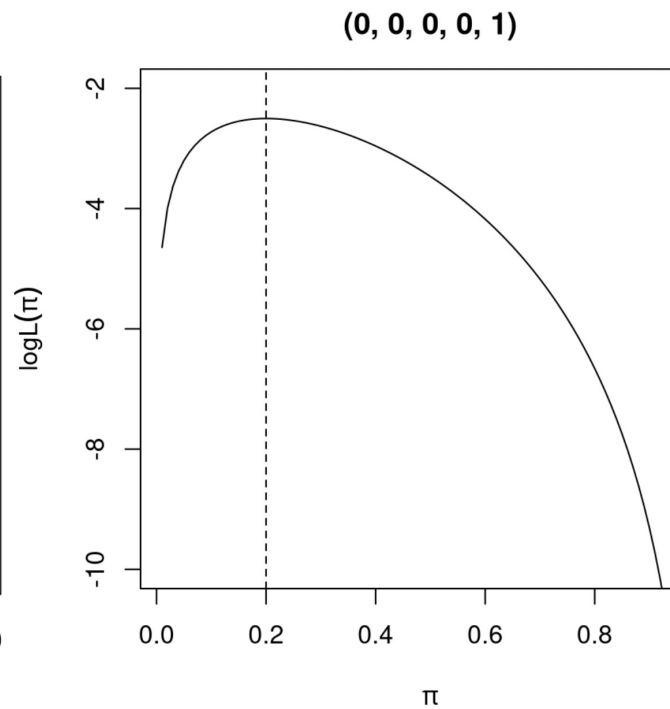
$$s(\theta; y_1, \dots, y_n) = \sum_{i=1}^n s(\theta; y_i)$$

For details see: <https://discdown.org/microeconometrics/maximum-likelihood-estimation-1.html>

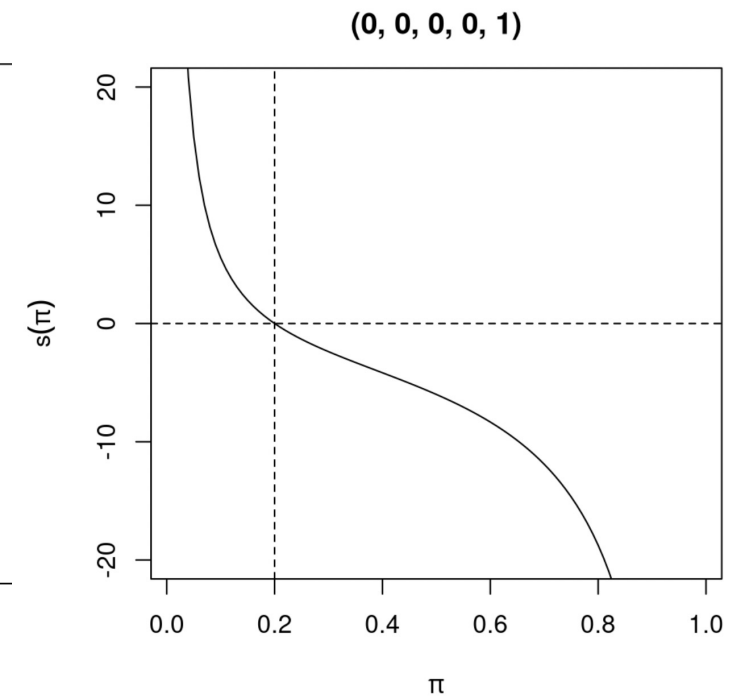
# Example: Bernoulli experiment



$$L(\pi; y) = \prod_{i=1}^n \pi^{y_i} (1 - \pi)^{1-y_i}$$



$$\ell(\pi; y) = \sum_{i=1}^n (1 - y_i) \log(1 - \pi) + y_i \log \pi$$



$$s(\pi; y) = \sum_{i=1}^n \frac{y_i - \pi}{\pi(1 - \pi)}$$

# MLE for ERGMs

$$\mathbb{P}_{\theta}(G) = \frac{\exp(\theta^T \cdot \Gamma(G))}{\sum_{G^* \in \mathcal{G}(N)} \exp(\theta^T \cdot \Gamma(G^*))}$$

$$c(\theta) := \sum_{G^* \in \mathcal{G}(N)} \exp(\theta^T \cdot \Gamma(G^*))$$

$$\mathbb{P}_{\theta}(G) \approx \prod_{ij} \mathbb{P}_{\theta}(g_{ij} | G_{-ij}),$$

$$\mathbb{P}_{\theta}(g_{ij} = 1 | G_{-ij}) = \text{logit}^{-1}(\theta^T \cdot \delta(\Gamma(G))),$$

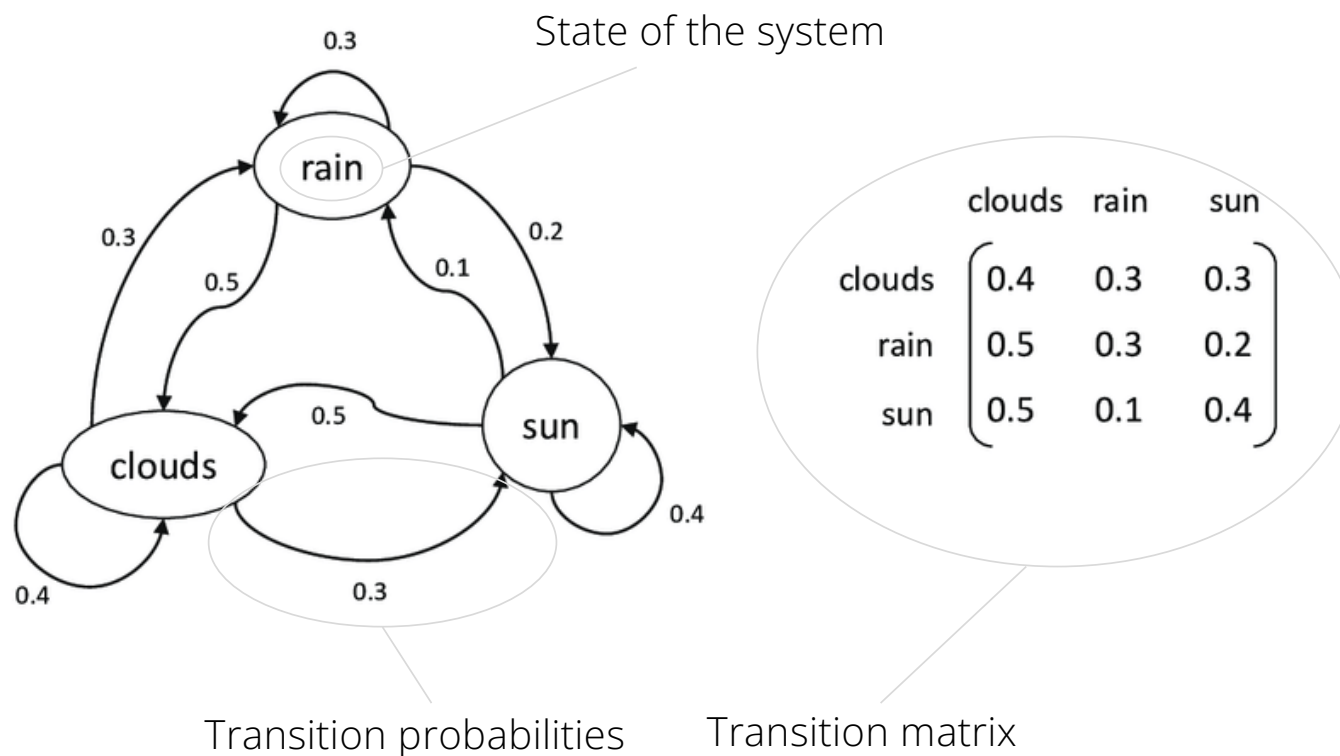
For details, see: <https://arxiv.org/pdf/1708.02598>

$$\begin{aligned} & \text{loglik}(\theta) - \text{loglik}(\theta_0) \\ &= -\log(c(\theta)) + \log(c(\theta_0)) \\ &= -\log\left(\frac{c(\theta)}{c(\theta_0)}\right) \\ &= -\log\left(\mathbb{E}_{\theta_0}\left[\exp\left((\theta - \theta_0)^T \cdot \Gamma(Y)\right)\right]\right) \\ &\approx -\log\left(\frac{1}{L} \cdot \sum_{i=1}^L \exp\left((\theta - \theta_0)^T \cdot \Gamma(G_i)\right)\right). \end{aligned}$$

By differentiating this equation on both sides with respect to  $\theta$  one gets an approximate score function:

$$s(\theta) \approx -\frac{\partial}{\partial \theta} \log\left(\frac{1}{L} \cdot \sum_{i=1}^L \exp\left((\theta - \theta_0)^T \cdot \Gamma(G_i)\right)\right) \quad (2)$$

# Markov Chains



A **stationary distribution** of a **Markov chain** is a probability distribution that remains unchanged in the Markov chain as time progresses.

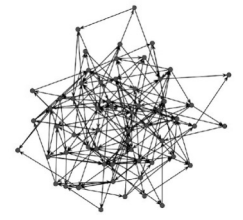
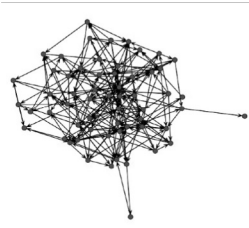
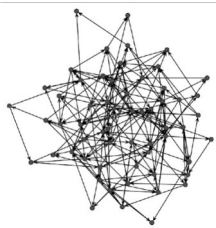
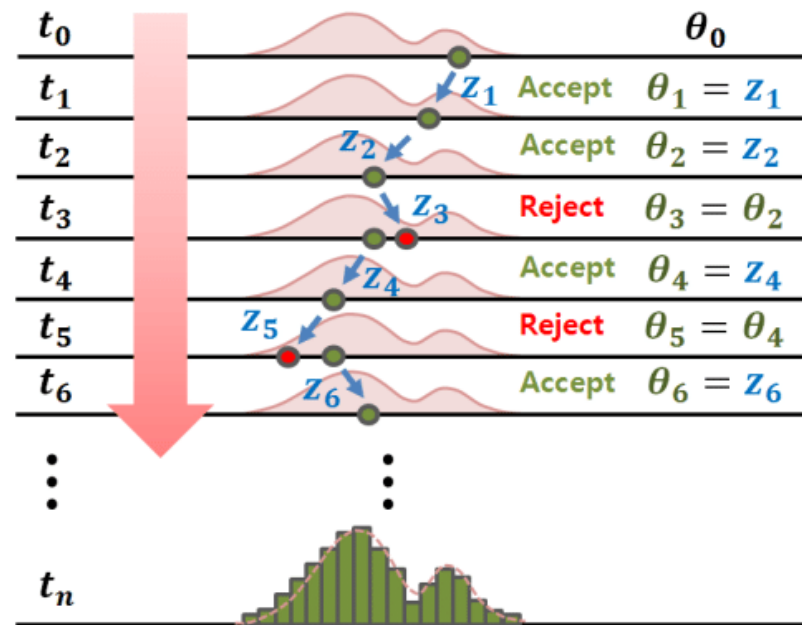
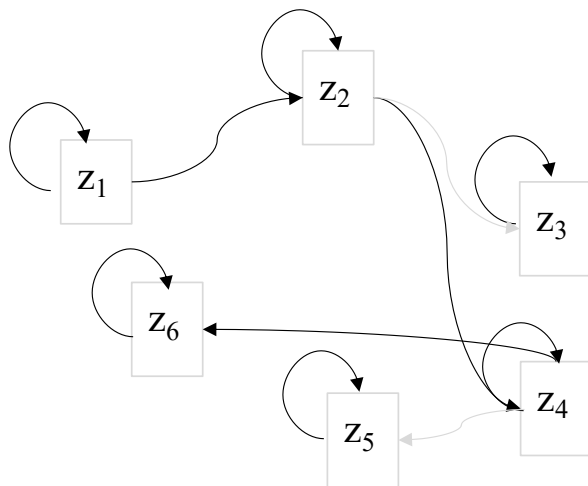
Typically, it is represented as a row vector  $\pi$  whose entries are probabilities summing to 1, and given **transition matrix**  $\mathbf{P}$ , it satisfies

$$\pi = \pi \mathbf{P}.$$

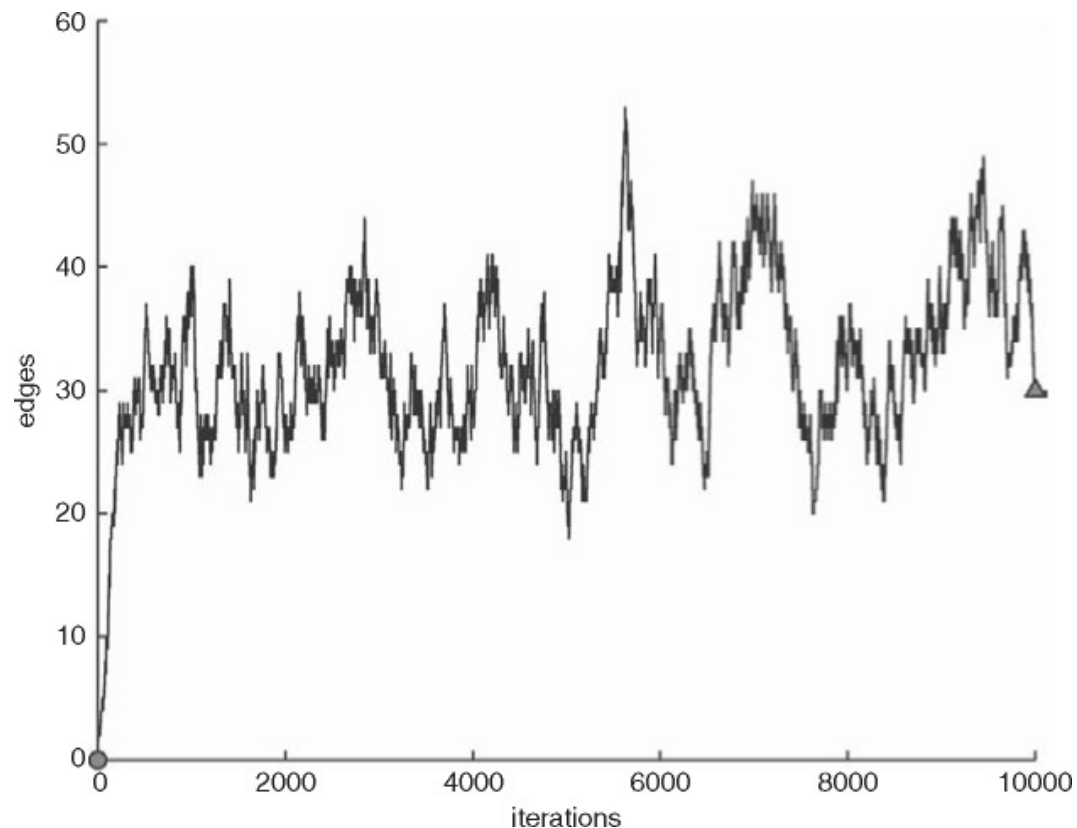
In other words,  $\pi$  is *invariant* by the matrix  $\mathbf{P}$ .

For details, see Moore, W. H., & Siegel, D. A. (2013). *A mathematics course for political and social research*. Princeton University Press. Section IV, Part 14

# Monte Carlo Markov Chains (MCMC)



## Exapmle MCMC for edge count



For details see: Lusher, D., Koskinen, J., & Robins, G. (Eds.). (2013). *Exponential random graph models for social networks: Theory, methods, and applications*. Cambridge University Press. Chapter 12 &13

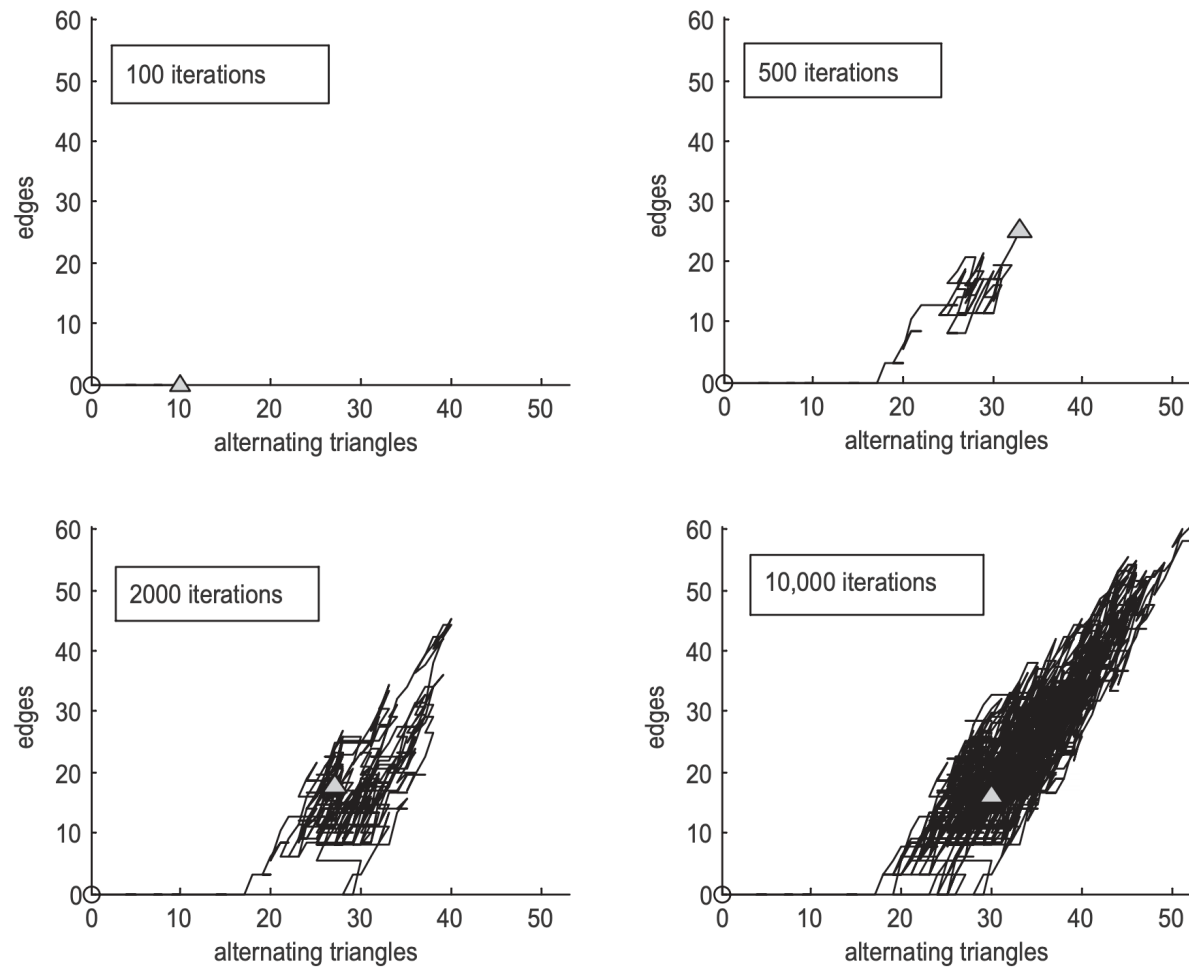
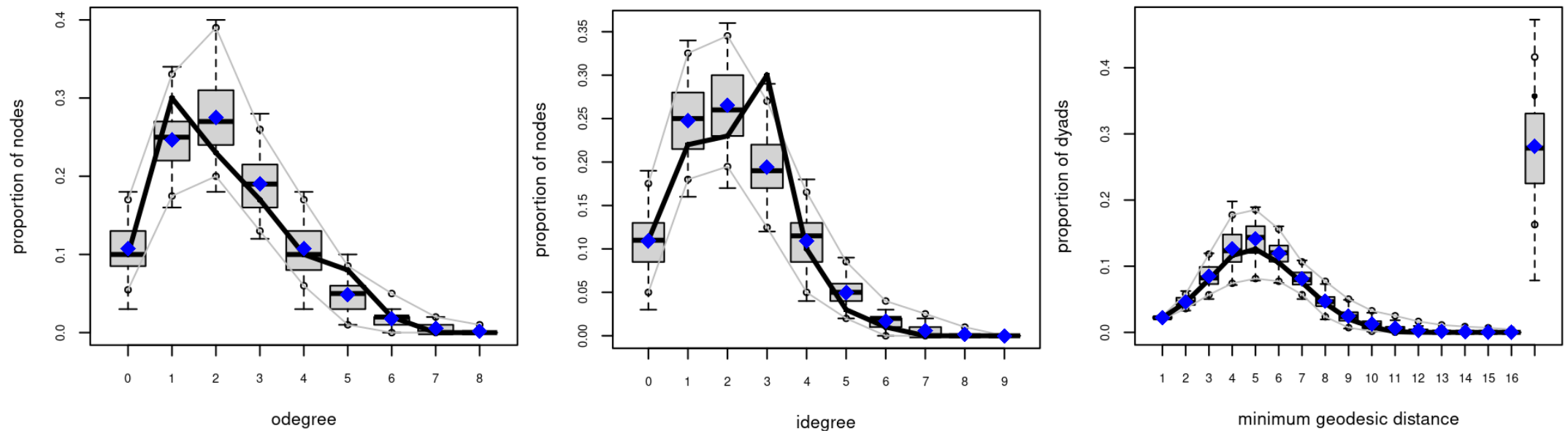


Figure 12.3. Number of edges against alternating triangles for sequence of graphs in Markov chain starting in empty graph. Model with edge and alternating triangle parameters equal to  $-2$  and  $0.3$  respectively,  $n = 20$ .

# Model diagnostics: Goodness of fit (GOF) example



For details see: Lusher, D., Koskinen, J., & Robins, G. (Eds.). (2013). *Exponential random graph models for social networks: Theory, methods, and applications*. Cambridge University Press. Chapter 12&13

# Summary

- ERGMs allow to model tie formation processes producing the global structure of empirical networks
- Interpretation of parameters similar to logOdds in logistic regression models
- Typical way to obtain parameters is approximation of maximum likelihood via MCMC
- Goodness of fit is assessed by simulating networks from an estimated ERGM and checking whether network properties of simulated networks are similar to empirical network

# Organizational details

- Next week will be half discussion, half lab session. Bring your **laptops** and pre-install **R** and **R Studio**

# Readings for next week

Wittek, M., Bartenhagen, C., & Berthold, F. (2023). The development of stratification and segregation in a new scientific field: A study of collaboration among scientists in neuroblastoma research between 1975 and 2016. ***Social Networks***, **72**, 80-107.

## Optional but highly recommended:

Luke, D. A. (2015). ***A user's guide to network analysis in R*** (Vol. 72, No. 10.1007, pp. 978-3). Cham: Springer.

**Thank you for your attention!**